

```
DataScience...yomide (611)*
181  /* Extending analysis using JOINS, SUBQUERIES, and Common
182  Table Expressions (CTEs)
183
184  Writing queries using JOINS
185
186  Retrieving a dataset that shows the sale_id,
187  product_name, customer_name, employee_name, quantity,
188  and sale_date by combining the relevant tables */
189  SELECT
190      s.sale_id,
191      p.product_name,
192      c.customer_name,
193      e.employee_name,
```

sale_id	product_name	customer_name	employee_name	quantity	sale_date
1	S101	Indomie Chicken Noodles	Emeka Obi	Adevala Johnson	2025-01-08

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (611) QuickMart 00:00:00 Row: 1, Col: 1 | 20 rows

```
DataScience...t connected*
187  product_name, customer_name, employee_name, quantity,
188  and sale_date by combining the relevant tables */
189  SELECT
190      s.sale_id,
191      p.product_name,
192      c.customer_name,
193      e.employee_name,
194      s.quantity,
195      s.sale_date
196  FROM Sales s
197  JOIN Products p ON s.product_id = p.product_id
198  JOIN Customers c ON s.customer_id = c.customer_id
199  JOIN Employees e ON s.employee_id = e.employee_id
```

sale_id	product_name	customer_name	employee_name	quantity	sale_date
1	S101	Indomie Chicken Noodles	Emeka Obi	Adevala Johnson	2025-01-08

Disconnected.

```
DataScience...t connected*
196  FROM Sales s
197  JOIN Products p ON s.product_id = p.product_id
```

sale_id	product_name	customer_name	employee_name	quantity	sale_date
1	S101	Indomie Chicken Noodles	Emeka Obi	Adevala Johnson	2025-01-08
2	S102	Coca-Cola 50cl	Fatima Abdull...	Chioma Okafor	2025-01-10
3	S103	Golden Penny Spaghetti	David Ajayi	Daniel Williams	2025-01-14
4	S104	Ariel Detergent 1kg	Ether williams	Grace Nwosu	2025-01-18
5	S105	Dangote Sugar 1kg	Yusuf Ibrahim	Adevala Johnson	2025-01-22
6	S106	Peak Milk 400g	Mary Johnson	Chioma Okafor	2025-02-03
7	S107	Kings Vegetable Oil 1L	Chinedu Okoro	Daniel Williams	2025-02-07
8	S108	Eva Water 1.5L	Tolulope Ade...	Grace Nwosu	2025-02-12
9	S109	Closeup Toothpaste 140g	Sarah Eze	Adevala Johnson	2025-02-18
10	S110	Milo 500g	Abdulahman ...	Chioma Okafor	2025-02-25
11	S111	Indomie Onion Noodles	Kemi Afolabi	Daniel Williams	2025-03-02
12	S112	Omo Detergent 1kg	Michael Peters	Grace Nwosu	2025-03-09
13	S113	Dettol Antiseptic 250ml	Emeka Obi	Adevala Johnson	2025-03-15
14	S114	Golden Morn 500g	Fatima Abdull...	Chioma Okafor	2025-03-21
15	S115	Peak Evaporated Milk	David Ajayi	Daniel Williams	2025-03-28
16	S116	Indomie Chicken Noodles	Ether williams	Grace Nwosu	2025-04-04
17	S117	Coca-Cola 50cl	Yusuf Ibrahim	Adevala Johnson	2025-04-11
18	S118	Kings Vegetable Oil 1L	Mary Johnson	Chioma Okafor	2025-04-19
19	S119	Ariel Detergent 1kg	Chinedu Okoro	Daniel Williams	2025-04-25
20	S120	Golden Penny Spaghetti	Tolulope Ade...	Grace Nwosu	2025-05-03

Disconnected.

DataScience...yomide (63))

```

201  /* Displaying each product along with the total
202  quantity sold */
203  SELECT product_name, SUM(quantity) AS total_quantity
204  FROM Sales s
205  JOIN Products p ON s.product_id = p.product_id
206  GROUP BY product_name

```

Ln: 206, Ch: 22 (205 chars, 6 lines) SPC CRLF UTF-8

Results Messages

product_name	total_quantity
1 Ariel Detergent 1kg	9
2 Closeup Toothpa...	2
3 Coca-Cola 50cl	12
4 Dangote Sugar 1kg	6
5 Dettol Antiseptic ...	2
6 Eva Water 1.5L	10
7 Golden Mom 500g	4
8 Golden Penny Sp...	6
9 Indomie Chicken ...	11
10 Indomie Onion N...	5
11 Kings Vegetable ...	5
12 Milo 500g	1

Query executed successfully. AYOMIDE-PC\SQL EXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 15 rows

DataScience...yomide (63))

```

202  quantity sold */
203  SELECT product_name, SUM(quantity) AS total_quantity
204  FROM Sales s
205  JOIN Products p ON s.product_id = p.product_id
206  GROUP BY product_name

```

Ln: 206, Ch: 22 (205 chars, 6 lines) SPC CRLF UTF-8

Results Messages

product_name	total_quantity
1 Ariel Detergent 1kg	9
2 Closeup Toothpa...	2
3 Coca-Cola 50cl	12
4 Dangote Sugar 1kg	6
5 Dettol Antiseptic ...	2
6 Eva Water 1.5L	10
7 Golden Mom 500g	4
8 Golden Penny Sp...	6
9 Indomie Chicken ...	11
10 Indomie Onion N...	5
11 Kings Vegetable ...	5
12 Milo 500g	1
13 Ono Detergent 1kg	3
14 Peak Evaporated...	6
15 Peak Milk 400g	2

Query executed successfully. AYOMIDE-PC\SQL EXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 15 rows

DataScience...yomide (62))

```

208  /* Displaying each employee along with the number of
209  sales they have handled */
210  SELECT employee_name, COUNT(quantity) AS number_of_sale
211  FROM Sales s
212  JOIN Employees e ON s.employee_id = e.employee_id
213  GROUP BY employee_name
214
215  /* Displaying each customers alongside the total
216  number of transactions they have made */

```

Ln: 208, Ch: 1 (228 chars, 6 lines) SPC CRLF UTF-8

Results Messages

employee_name	number_of_sales
1 Adewale Johnson	5
2 Chioma Okafor	5
3 Daniel Williams	5
4 Grace Nwosu	5

Query executed successfully. AYOMIDE-PC\SQL EXPRESS (17.0... Ayomide-PC\Ayomide (62) QuickMart 00:00:00 Row: 1, Col: 1 4 rows

DataScience...yomide (62))

```

215  /* Displaying each customers alongside the total
216  number of transactions they have made */
217  SELECT customer_name, COUNT(quantity) AS total_transact
218  FROM Sales s
219  JOIN Customers c ON s.customer_id = c.customer_id
220  GROUP BY customer_name

```

Results

	customer_name	total_transactions
1	Abdurahman Bello	1
2	Chinedu Okoro	2
3	David Ayayi	2
4	Emeka Obi	2
5	Eather williams	2
6	Fatma Abdullahi	2
7	Kemi Afolabi	1
8	Mary Johnson	2
9	Michael Peters	1
10	Sarah Eze	1
11	Tolope Adekunle	2
12	Yusuf Ibrahim	2

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (62) QuickMart 00:00:00 Row: 1, Col: 1 12 rows

DataScience...yomide (63))

```

222  /* Using subqueries
223
224  Retrieving products whose price is above the average
225  product price */
226  SELECT product_name, price
227  FROM Products
228  WHERE price > (
229      SELECT AVG(price) FROM Products
230  )

```

Results

	product_name	price
1	Peak Milk 400g	3200.00
2	Milo 500g	4500.00
3	Kings Vegetable Oil 1L	3200.00
4	Dettol Antiseptic 250ml	2800.00
5	Ariel Detergent 1kg	4200.00
6	Golden Mom 500g	2300.00
7	Omo Detergent 1kg	3800.00

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 7 rows

DataScience...yomide (63))

```

232  /* Identifying employees whose salary is above the
233  average salary */
234  SELECT employee_name, salary
235  FROM Employees
236  WHERE salary > (
237      SELECT AVG(salary) FROM Employees
238  )
239
240  -- Retrieving product(s) with the highest price

```

Results

	employee_name	salary
1	Ibrahim Musa	275000
2	Blessing Eze	320000
3	Tunde Adeyemi	300000
4	Samuel Peters	290000
5	Funke Adebayo	340000

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 5 rows

```
DataScience...yomide (62)*
239
240 -- Retrieving product(s) with the highest price
241 SELECT product_name, price
242 FROM Products
243 WHERE price = (
244     SELECT MAX(price) FROM Products
245 )
246
247 /* Identifying customers who have made purchases with
248 quantities greater than the average quantity per
249 transaction */
250
```

Ln: 240, Ch: 1 (147 chars, 6 lines) SPC CRLF UTF-8

product_name	price
Milo 500g	4500.00

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (62) QuickMart 00:00:00 Row: 1, Col: 1 1 rows

```
DataScience...yomide (63)*
247 /* Identifying customers who have made purchases with
248 quantities greater than the average quantity per
249 transaction */
250 SELECT customer_name, quantity
251 FROM Sales s
252 JOIN Customers c ON s.customer_id = c.customer_id
253 WHERE quantity > (
254     SELECT AVG(quantity) FROM Sales
255 )
256
```

Ln: 255, Ch: 2 (277 chars, 9 lines) SPC CRLF UTF-8

customer_name	quantity
1 Fatima Abdullahi	5
2 Yusuf Ibrahim	6
3 Tolulope Adekunle	10
4 Kemi Afolabi	5
5 David Ajayi	6
6 Esther Williams	8
7 Yusuf Ibrahim	7
8 Chinedu Okoro	5

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 8 rows

```
DataScience...yomide (63)*
257 /* Using Common Table Expressions (CTEs)
258
259 Creating a CTE that lists total quantity sold per
260 product and using it to retrieve products with the
261 highest sales volume */
262 WITH ProductSales AS (
263     SELECT
264         product_id,
265         SUM(quantity) AS total_sold
266     FROM Sales
267     GROUP BY product_id
268 )
269 SELECT
270     p.product_name
271
```

Ln: 274, Ch: 28 (459 chars, 18 lines) SPC CRLF UTF-8

product_name	total sold
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Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 15 rows

```
DataScience...yomide (63)*
262 WITH ProductSales AS (
263     SELECT
264         product_id,
265         SUM(quantity) AS total_sold
266     FROM Sales
267     GROUP BY product_id
268 )
269 SELECT
270     p.product_name,
271     ps.total_sold
272 FROM ProductSales ps
273 JOIN Products p ON ps.product_id = p.product_id
274 ORDER BY ps.total_sold DESC
275
```

200 % 4 0 ↑ ↓ Ln: 274, Ch: 28 (459 chars, 18 lines) SPC CRLF UTF-8

Results Messages

product_name	total_sold
--------------	------------

Query executed successfully. AYOMIDE-PC\SQL EXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 15 rows

```
DataScience...yomide (63)*
270     p.product_name,
271     ps.total_sold
272 FROM ProductSales ps
273 JOIN Products p ON ps.product_id = p.product_id
274 ORDER BY ps.total_sold DESC
```

200 % 4 0 ↑ ↓ Ln: 274, Ch: 28 (459 chars, 18 lines) SPC CRLF UTF-8

Results Messages

product_name	total_sold
1 Coca-Cola 50cl	12
2 Indomie Chicken Noodles	11
3 Eva Water 1.5L	10
4 Ariel Detergent 1kg	9
5 Peak Evaporated Milk	6
6 Golden Penny Spaghetti	6
7 Dangote Sugar 1kg	6
8 Kings Vegetable Oil 1L	5
9 Indomie Onion Noodles	5
10 Golden Mom 500g	4
11 Omo Detergent 1kg	3
12 Dettol Antiseptic 250ml	2
13 Closeup Toothpaste 140g	2
14 Peak Milk 400g	2
15 Milo 500g	1

Query executed successfully. AYOMIDE-PC\SQL EXPRESS (17.0... Ayomide-PC\Ayomide (63) QuickMart 00:00:00 Row: 1, Col: 1 15 rows

```
DataScience...yomide (62)*
276 /* Creating another CTE that summarizes the number of
277 sales handled by each employee and using it to identify
278 the employee with the highest number of sales */
279 WITH EmployeeSales AS (
280     SELECT
281         employee_id,
282         COUNT(quantity) AS sales
283     FROM Sales
284     GROUP BY employee_id
285 )
286 SELECT
```

200 % 4 0 ↑ ↓ Ln: 276, Ch: 1 (441 chars, 16 lines) SPC CRLF UTF-8

Results Messages

employee_name	sales
1 Adewale Johnson	5
2 Chioma Okafor	5
3 Daniel Williams	5
4 Grace Nwosu	5

Query executed successfully. AYOMIDE-PC\SQL EXPRESS (17.0... Ayomide-PC\Ayomide (62) QuickMart 00:00:00 Row: 1, Col: 1 4 rows

DataScience...yomide (62)*

```

282     COUNT(quantity) AS sales
283 FROM Sales
284 GROUP BY employee_id
285 )
286 SELECT
287     e.employee_name,
288     es.sales
289 FROM EmployeeSales es
290 JOIN Employees e ON es.employee_id = e.employee_id
291 ORDER BY es.sales DESC
292

```

200 % 4 0 0 0 Ln: 276, Ch: 1 (441 chars, 16 lines) SPC CRLF UTF-8

	employee_name	sales
1	Adele Johnson	5
2	Choma Okafor	5
3	Daniel Williams	5
4	Grace Nwosu	5

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (62) QuickMart 00:00:00 Row: 1, Col: 1 4 rows

DataScience...yomide (62)*

```

293 /* Creating a CTE that lists customers and their total
294 transactions, and retrieving customers with more than
295 three transactions */
296 WITH CustomerTransactions AS (
297     SELECT
298         customer_id,
299         COUNT(quantity) AS total_transactions
300     FROM Sales
301     GROUP BY customer_id
302 )
303 SELECT

```

200 % 4 0 0 0 Ln: 293, Ch: 1 (464 chars, 16 lines) SPC CRLF UTF-8

customer_name	total_transactions
---------------	--------------------

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (62) QuickMart 00:00:00 Row: 1, Col: 1 0 rows

DataScience...yomide (62)*

```

299     COUNT(quantity) AS total_transactions
300 FROM Sales
301 GROUP BY customer_id
302 )
303 SELECT
304     c.customer_name,
305     ct.total_transactions
306 FROM CustomerTransactions ct
307 JOIN Customers c ON ct.customer_id = c.customer_id
308 WHERE ct.total_transactions > 3

```

200 % 4 0 0 0 Ln: 293, Ch: 1 (464 chars, 16 lines) SPC CRLF UTF-8

customer_name	total_transactions
---------------	--------------------

Query executed successfully. AYOMIDE-PC\SQLEXPRESS (17.0... Ayomide-PC\Ayomide (62) QuickMart 00:00:00 Row: 1, Col: 1 0 rows